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Appendix

Roadmap. In Section A, we introduce the fixed point method and the well-known Banach fixed-point theorem. In Section B, we provide the analysis of fixed points under the polynomial activation function. In Section C, we provide the analysis of fixed points under the exponential activation function.

A TOOLS OF FIXED POINT METHODS

In this section, we present the tools of fixed point methods used in our work. In Section A.1, we introduce the general case of Banach fixed point theorem. In Section A.2, we present the scalar case of Banach fixed point theorem. In Section A.3, we present the vector case of Banach fixed point theorem. In Section A.4, we present the matrix case of Banach fixed point theorem. In Section A.5, we present some calculation examples using Banach fixed point theorem.

A.1 Banach Fixed Point Theorem

We firstly present the definition of fixed point.

Definition A.1 (Fixed point, [AH09]). *Let V be a Banach space with the norm $\|\cdot\|_V$, and let D be a subset of V . We say a function $f : D \rightarrow V$ has a fixed point $p \in D$ if $f(p) = p$.*

And we introduce the concept of contractive mapping which will be used later.

Definition A.2 (Contractive mapping, Definition 5.1.2 of [AH09]). *Let V be a Banach space with the norm $\|\cdot\|_V$, and let D be a subset of V . We say that a function $f : D \rightarrow V$ is contractive with contractivity constant $K \in [0, 1)$ if*

$$\|f(x) - f(x')\|_V \leq K\|x - x'\|_V, \quad \forall x, x' \in D.$$

Now, we can present the Banach fixed point theorem, which is crucial in guaranteeing the existence and uniqueness of fixed points in metric spaces under contractive mappings, forming the foundation of many applications in analysis and applied mathematics.

Lemma A.3 (Banach fixed point theorem, Theorem 5.1.3 of [AH09]). *Let V be a Banach space V with the norm $\|\cdot\|_V$. Let D be a nonempty closed set of V . Suppose that $f : D \rightarrow V$ is a mapping that satisfies the following*

- $f(x) \in D$ whenever $x \in D$.
- f is contractive with contractivity constant $K \in [0, 1)$.

Then it holds that

- The function f has a unique fixed point $p \in D$.
- For any initial point $x^{(0)} \in D$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.

- The following error bounds hold:

$$\begin{aligned}\|x^{(t)} - p\|_V &\leq \frac{K^t}{1-K} \|x^{(1)} - x^{(0)}\|_V, \\ \|x^{(t)} - p\|_V &\leq \frac{K}{1-K} \|x^{(t)} - x^{(t-1)}\|_V, \\ \|x^{(t)} - p\|_V &\leq K \|x^{(t-1)} - p\|_V.\end{aligned}$$

A.2 Scalar Case

In this section, we present the scalar case of Banach fixed point theorem.

Lemma A.4 (Banach fixed point theorem, scalar case). *Let $D \subseteq \mathbb{R}$ be a nonempty closed set. Suppose that $f : D \rightarrow \mathbb{R}$ is differentiable and satisfies the following:*

- $f(x) \in D$ whenever $x \in D$.
- There exists constant $K < 1$ such that

$$|f'(x)| \leq K, \quad \forall x \in D.$$

Then it holds that

- The function f has a unique fixed point $p \in D$.
- For any initial point $x^{(0)} \in D$, the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.
- The following error bounds hold:

$$\begin{aligned}|x^{(t)} - p| &\leq \frac{K^t}{1-K} |x^{(1)} - x^{(0)}|, \\ |x^{(t)} - p| &\leq \frac{K}{1-K} |x^{(t)} - x^{(t-1)}|, \\ |x^{(t)} - p| &\leq K |x^{(t-1)} - p|.\end{aligned}$$

Proof. It directly holds by Lemma A.5 with $d = 1$. □

A.3 Vector Case

In this section, we extend the Banach fixed point theorem to its vector version. Note that $\mathbb{R}^d$ is a Banach space with norm $\|\cdot\|_\infty$. The ℓ_∞ norm of $x \in \mathbb{R}^d$ is defined as $\|x\|_\infty = \max_{i \in [d]} |x_i|$.

Lemma A.5 (Banach fixed point theorem, vector case, formal version of Lemma 3.4). *Let $D \subseteq \mathbb{R}^d$ be a nonempty closed set. Suppose that $f : D \rightarrow \mathbb{R}^d$ is differentiable and satisfies the following:*

- $f(x) \in D$ whenever $x \in D$.
- There exists constant $K < 1$ such that for every $i \in [d]$,

$$\left\| \frac{df_i(x)}{dx} \right\|_1 \leq K \quad \forall x \in D.$$